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Department of Software Engineering
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Software Architectures

Practice 2

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Project work

- List of projects: Project Work
- Submission deadline: May 10, 2026, 23:59
- A topic that cannot be picked:
  EventHub - Community Event Platform (*A platform where users can create, browse, and register for local events. Organizers can manage attendees, send updates, and collect feedback after events.*)

Functional requirements vs. Quality Attributes

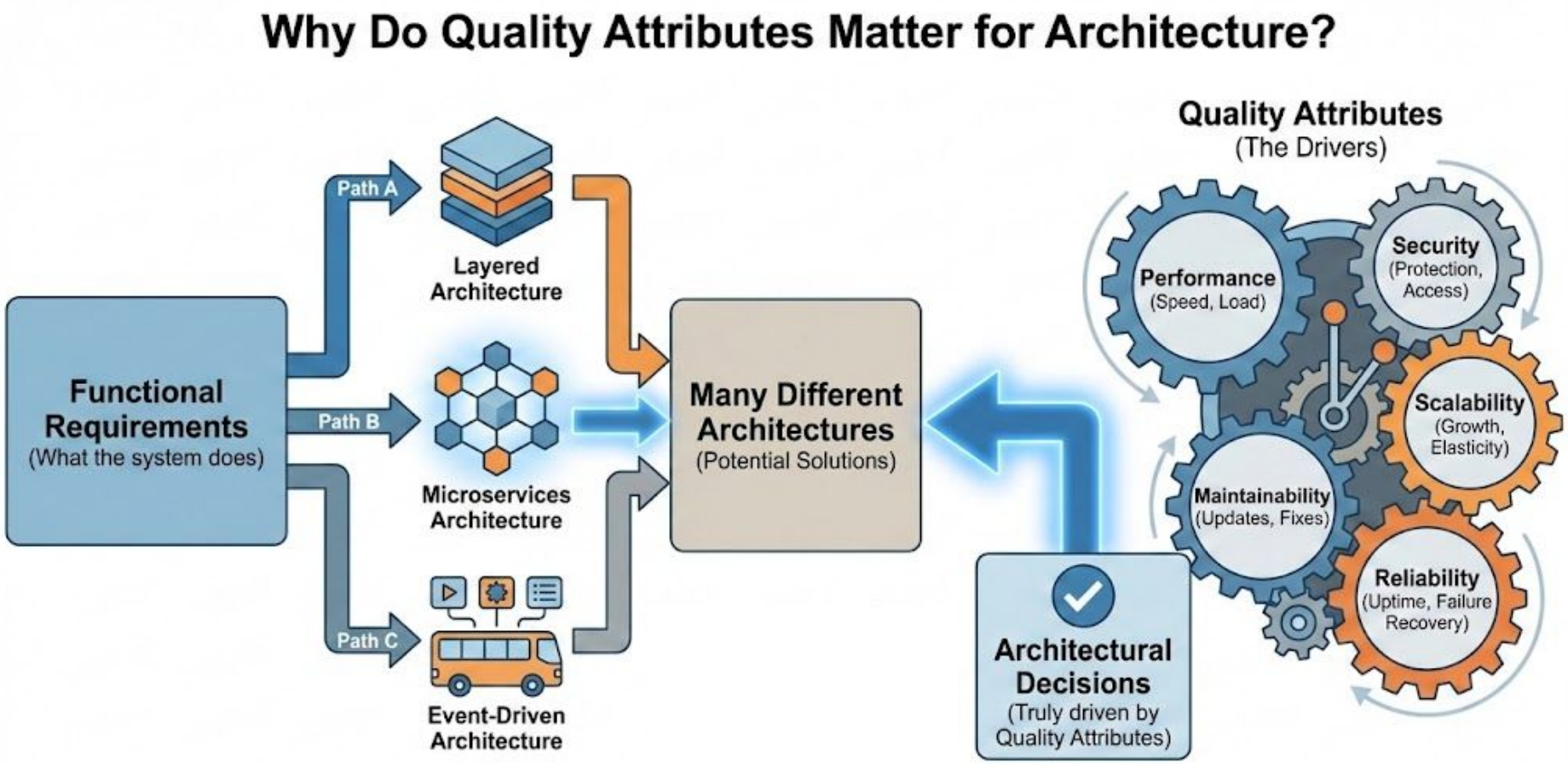
Example 1: "The system shall allow customers to book tickets."

Architecture 1: ???

Example 2: “The system shall allow customers to book tickets with sub-second response time (performance), even when 10,000 users are booking simultaneously (scalability), ensuring no ticket is sold twice (consistency/reliability).”

Architecture 2: ???

Importance of Quality Attributes



Quality attributes act as the primary filters, selecting the specific architecture that best meets non-functional goals, even when functional requirements allow for multiple options.

What are Quality Attributes?

"-ilities"

Quality Attribute	Question it answers
Performance	How fast does the system respond?
Scalability	Can it handle growing numbers of users?
Availability	What percentage of time is the system operational?
Security	How well does it protect data and prevent unauthorized access?
Modifiability	How easy is it to change or extend the system?
Testability	How easy is it to verify the system works correctly?
Usability	How easy is the system for end users?
Deployability	How easy is it to release new versions?

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How to answer Quality Attribute related questions?

- Define Quality Attribute Scenarios!

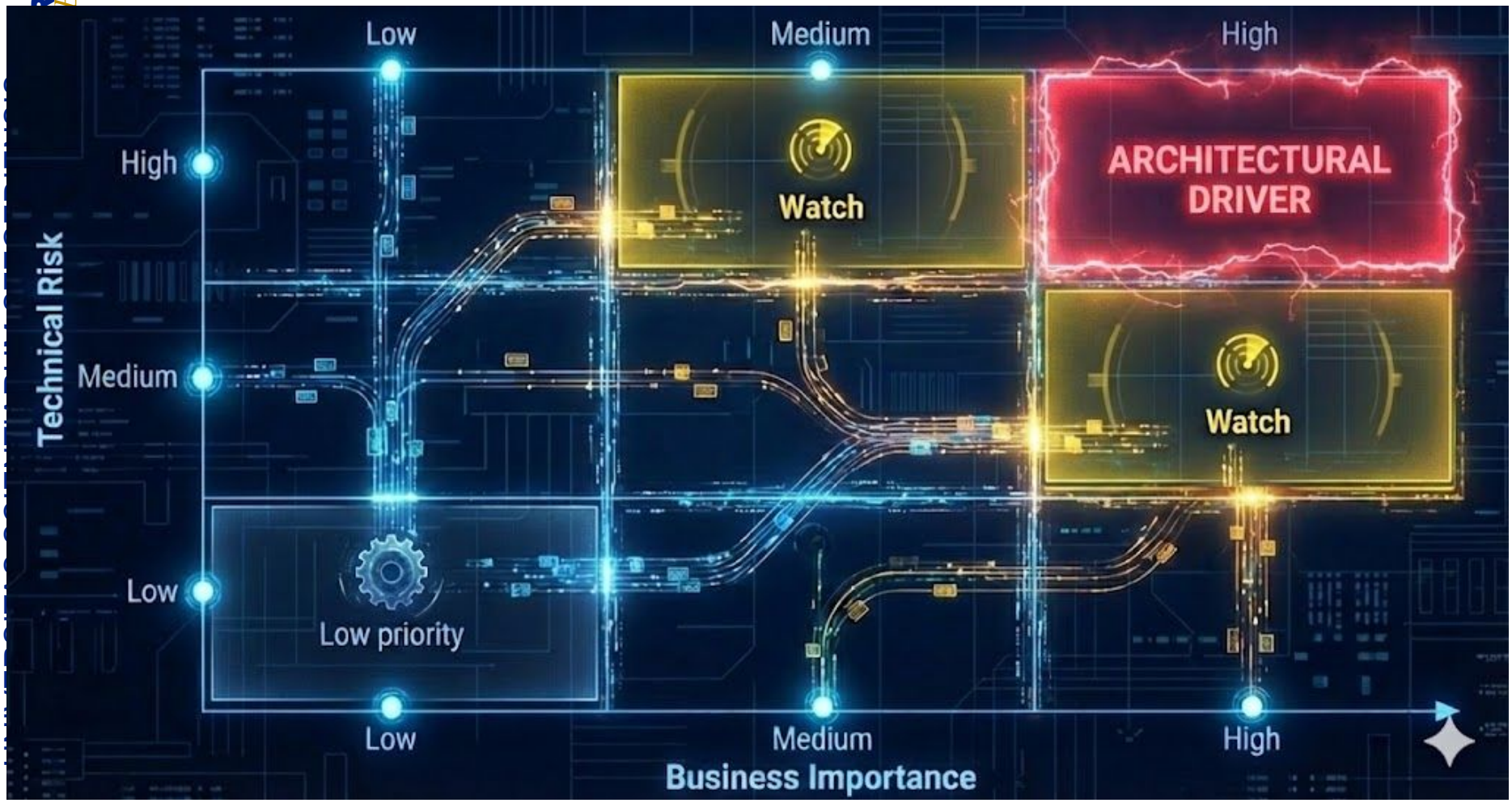
Part	Description	Example
Source	Who or what triggers the scenario?	10,000 concurrent users
Stimulus	What happens?	Submit ticket purchase requests
Artifact	What part of the system is affected?	Booking service
Environment	Under what conditions?	Normal operation, peak load
Response	What does the system do?	Processes all requests, no double-booking
Response Measure	How do we measure success?	99.9% of requests complete in < 2 seconds

Quality Attribute Workshop (QAW)

1. Brainstorm Quality Attribute Scenarios
2. Consolidate and refine
3. Prioritize
4. Document

developed by Software Engineering Institute (SEI)

Prioritization Grid



The Decision Matrix

- A tool for transparent, justifiable choices between alternatives
  - evaluate options against criteria of your project
- 4 reasons:
  - Transparency: everyone can see why a decision was made
  - Objectivity: reduces bias toward familiar technologies
  - Documentation: Creates a record that future team members can understand
  - Justifiability: When someone asks "why did you choose X?", you have a clear answer

Building a Decision Matrix

1. Define the decision!
   a. be specific, e.g.: “choose a backend framework for project X”, and not “choose a technology”
2. Identify criteria!
   a. use your quality attribute priorities
3. Assign weights!
   a. distribute 100 points across criteria based on your QAW results
4. Identify alternatives!
5. Score each alternative!
6. Calculate weighted scores!

The process

1. Specification - WHAT?
   a. User stories
   b. Capabilities
   c. Roles
   d. Core features
2. Quality Attribute Workshop - HOW?
   a. Quality Attributes Scenarios
   b. Architectural Drivers
3. Decision Matrix - WITH WHAT?
   a. Weighted criteria
   b. Technology choices